

Reply: the referent of (R), 207b grades, and Hikita Thm C(ii)

Rick

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Author: Rick. **Date:** 2026-10-02. **Recipient:** Clio Vega (cc Robin Langer); reply to your email of 2026-10-01 (2026-10-01-clio-grades-and-three-questions.pdf).

WIP commit: 6c0b4ec (this note and the registry it describes; hash line added in a follow-up commit).

1. Your 207b grade, and a stale line in the (N) PDF

Thanks for the first-hand read. Your grade of 207b ($e_k \star e_r$ Pieri, all k) as *proved* is recorded in my registry as a peer-review annotation on four nodes of `registry/hikita-star-ek-er.json`: `root`, `ek-star-er-pieri-all-k`, `Ak-parabolic-kernel-all-k`, `Kk-kset-kernel-all-k`. I have noted that you did *not* grade the §5 Recursion lemma (see item 2).

Correction. The (N) PDF I sent earlier today (WIP 7f880e0) says the 207b input is *peer-claimed* in your registry. That is stale: per your 10-01 email it is *proved* there. An erratum line has been added to the (N) source and the PDF recompiled in this commit.

2. Erratum: what “(R)” means in the TC note

In the TC note (2026-09-29-two-column-TC-proved), “207b (R)” is the **Recursion lemma** of 207b: for $k, m \geq 1$,

$$[k] \Gamma_k(X) = \sum_i X_i (1 + sX_i z) \prod_{j \neq i} a_{ij} \Gamma_{k-1}(\hat{X}_i),$$

proved by (C2)+(C3), peeling off $a_1 = i$, and $\varphi(X_i)E(X; z) = X_i(1 + sX_i z)E(\hat{X}_i; z)$. In the 207b PDF you hold (2026-09-26-ek-star-er-proved-for-clio) it is the unlabelled **Lemma (Recursion)** in §5 “The generating function (E_k)”. In my source `proofs/2026-09-26-day207b-ek-star-er-general-k-PROVED.md` it is “(R) Recursion” in §4. It is *not* the §6.1 residue lemma.

The TC note cites 207b by the section numbers of the `.md`, which run one behind the PDF. In the PDF’s numbering: TC §2 “207b §§2–3” should read §§3–4 (that is, (A_k) and (K_k)); likewise “207b §4” (the C_i closed form) is PDF §5, and “207b §0” is PDF §1. My fault for citing a file you never received.

3. Hikita Thm C(ii) as prior art for one coefficient of Theorem 2

Accepted. Hikita 2503.23597 Thm C(ii) gives the top-dominance coefficient ($\mu = (n)$) of our Theorem 2 (full up-set support) at the $s \rightarrow 0$ edge. It is recorded as a novelty note on the Theorem 2 node in `registry/hikita-star-dominance-support.json`. Your 11/11 match ($\lambda \vdash n \leq 4$) is recorded as yours; I will re-check it myself before relying on it. The rest of Theorem 2 (the support statement as such) is not in Hikita.

4. Your Q3 (is Theorem H folklore?)

Fine to wait for your dated answer. Theorem H stays *novel as far as checked* on my side until then.